

Week 6

- 1.** Set up the development environment and initial backend structure for the AI avatar image generation service using Python and required dependencies.
- 2.** Implemented the main worker service (`avatar_image_service.py`) responsible for consuming jobs from the message queue and processing image generation requests.
- 3.** Designed an event-driven workflow where image generation requests are pushed to a queue and processed asynchronously by the worker service.
- 4.** Integrated the Google Gemini AI image model to generate different avatar styles such as Superhero, Gladiator, Pirate, Maharaja, and Rockstar.
- 5.** Implemented AWS S3 integration to download the base avatar images and upload the generated AI images after processing.
- 6.** Developed a Redis-based caching mechanism to store avatar files locally with an expiry time, reducing repeated downloads and improving performance.
- 7.** Performed initial testing using producer and consumer scripts to verify correct queue processing, job execution, and system workflow.